Cardiovascular Diabetology

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Joint longitudinal trajectories of the triglyceride-glucose index combined with BMI and waist-to-height ratio and incident cardiovascular disease: a prospective cohort study from the English longitudinal study of ageing

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Abstract

Background The triglyceride-glucose (TyG) index and its composite obesity indices have been linked to cardiovascular disease (CVD) risk. However, most prior studies relied on single baseline measurements, and few have employed group-based multi-trajectory modeling to capture concurrent longitudinal changes in metabolic and anthropometric indicators. This study aimed to identify joint longitudinal trajectory groups of TyG combined with body mass index (BMI) and waist-to-height ratio (WHtR) using parallel approaches and evaluate their associations with incident CVD in middle-aged and older adults.

Methods This prospective cohort study included 1808 CVD-free participants aged ≥ 50 years from the English Longitudinal Study of Ageing. Group-based multi-trajectory modeling was applied to jointly identify latent trajectory classes using repeated measurements of TyG with BMI and TyG with WHtR across three waves over approximately 8 years. Cox proportional hazards models with four sequential adjustment models estimated hazard ratios (HRs) and 95% confidence intervals (CIs) for incident composite CVD, heart disease, and stroke. Subgroup analyses were stratified by age, gender, smoking, diabetes, and hypertension status. Nine sensitivity analyses were conducted to assess robustness.

Results During follow-up, 263 participants (14.5%) developed incident CVD. Four trajectory groups were identified for each approach. In fully adjusted BMI + TyG models, compared with the normal weight–low TyG reference group,

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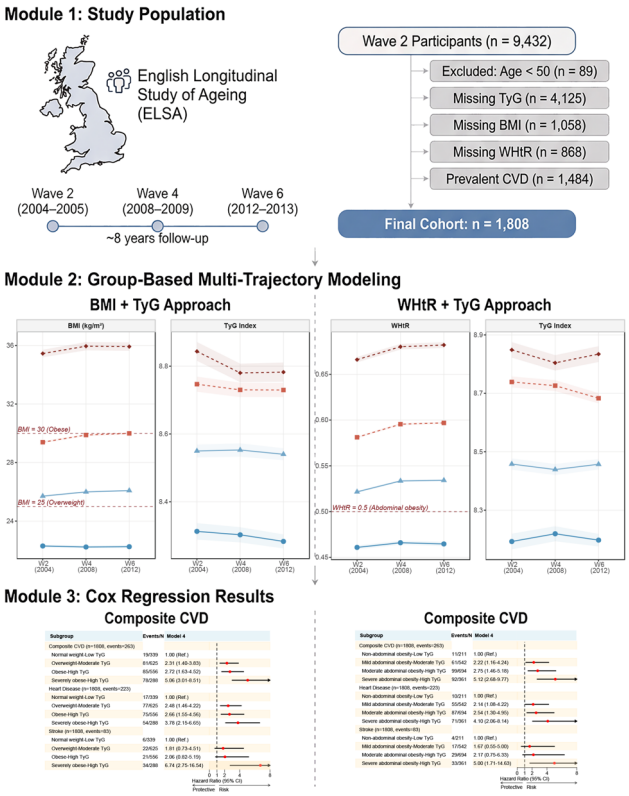
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composite CVD risk increased progressively across the overweight–moderate TyG (HR 2.31, 95% CI 1.40–3.83), obese–high TyG (HR 2.72, 95% CI 1.63–4.52), and severely obese–high TyG groups (HR 5.06, 95% CI 3.01–8.51). The WHtR + TyG approach demonstrated a consistent dose–response pattern, with HRs of 2.22, 2.75, and 5.12 for ascending risk groups. For stroke, only the highest-risk groups reached statistical significance (HR 6.74 and 5.00, respectively). Formal discriminative comparison showed no significant difference between the two approaches (C-statistic difference 0.003, $P = 0.723$). All nine sensitivity analyses consistently corroborated the primary findings.

Conclusions Both approaches yield robust and comparable dose–response gradients, supporting further validation of serial TyG-related composite index monitoring for cardiovascular risk stratification in aging populations.

Keyword Triglyceride-glucose index, Group-based multi-trajectory modeling, Cardiovascular disease, Obesity indicators, Longitudinal cohort study

Graphical abstract



Research Insights

What is currently known about this topic?

- The triglyceride-glucose (TyG) index and its composite obesity indices (TyG-BMI, TyG-WC, TyG-WHtR) are established surrogate markers of insulin resistance and have been associated with cardiovascular disease (CVD) risk across diverse populations.
- TyG-WC and TyG-WHtR generally demonstrate superior predictive performance for CVD compared with TyG-BMI in cross-sectional and single-timepoint cohort studies.
- Longitudinal trajectories of individual TyG-related indices provide better CVD prediction than single baseline measurements, but joint multi-trajectory approaches remain underexplored.

What is the key research question?

Are joint longitudinal trajectory groups of TyG combined with BMI versus WHtR differentially associated with incident CVD in middle-aged and older adults?

What is new?

- This is the first study to apply group-based multi-trajectory modeling to simultaneously identify joint

longitudinal trajectories of TyG with BMI and TyG with WHtR within the same cohort.
- Both approaches revealed strong dose–response associations with incident CVD, with hazard ratios reaching 5.06 and 5.12 in the highest-risk trajectory groups, respectively.
- Smoking status significantly modified these trajectory-CVD associations in both modeling approaches.

How might this study influence clinical practice?

Serial monitoring of TyG-related composite indices through joint trajectory assessment may enable earlier identification of high-risk individuals and support targeted cardiovascular prevention strategies in aging populations.

Introduction

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, accounting for approximately 19.2 million deaths in 2023 and imposing an enormous economic burden estimated at €282 billion annually in the European Union alone [1, 2]. Despite considerable advances in prevention and treatment, the global prevalence of CVD has more than doubled since 1990, driven largely by population aging and rising metabolic risk factor exposure, particularly elevated body mass index and fasting plasma glucose [2, 3]. In China, the CVD burden is especially pronounced, with a crude incidence rate of 620.33 per 100,000 reported in 2023, underscoring the urgent need for accessible and effective risk stratification tools [4].

Insulin resistance (IR) has been increasingly recognized as a pivotal contributor to atherosclerosis and CVD pathogenesis. The triglyceride-glucose (TyG) index, calculated from fasting triglycerides and glucose, has emerged as a convenient surrogate marker of IR and has demonstrated significant associations with CVD incidence and mortality across diverse populations [5, 6]. To further enhance predictive performance, researchers have combined the TyG index with anthropometric obesity indicators, generating composite indices such as TyG-BMI, TyG-WC, and TyG-WHtR. A recent meta-analysis of 85 studies confirmed that per 1-SD increase, these composite indices were associated with 17–20% higher CVD risk and 22–32% higher cardiovascular mortality risk, with TyG-WHtR and TyG-WC generally outperforming TyG-BMI [7]. Several cohort studies have further demonstrated that longitudinal trajectories of TyG-related indices, rather than single baseline measurements, provide superior predictive capacity for incident CVD [8–10]. Recent studies have also begun to compare multiple TyG-derived obesity indices simultaneously and to evaluate their longitudinal trajectories in relation to cardiovascular outcomes in prospective cohorts [11].

However, important gaps remain. While recent studies have advanced the field by examining longitudinal trajectories of individual TyG-related composite indices, most have modeled TyG and obesity indicators as pre-combined products (e.g., TyG-BMI as a single variable) rather than jointly modeling TyG and anthropometric measures as separate but concurrent trajectories, which may better capture the independent and synergistic dynamics of metabolic and adiposity changes over time. Furthermore, few studies have directly compared BMI-based and WHtR-based approaches within the same cohort using formal discriminative metrics, leaving it unclear whether these two anthropometric dimensions provide differential predictive value for cardiovascular events [12, 13]. Additionally, the existing evidence derives predominantly from East Asian populations, and data from European older adult cohorts remain limited [14, 15].

Therefore, the present study aimed to identify joint longitudinal trajectory groups of TyG combined with BMI and TyG combined with WHtR using group-based multi-trajectory modeling in a prospective cohort of middle-aged and older adults from the English Longitudinal Study of Ageing (ELSA). By jointly modeling TyG and anthropometric indicators as separate concurrent trajectories rather than using pre-combined composite indices, this approach allows the identification of latent subgroups defined by distinct patterns of metabolic and adiposity change. We sought to compare these two parallel approaches using formal discriminative metrics and evaluate their associations with incident composite CVD, heart disease, and stroke through Cox proportional hazards regression, with comprehensive subgroup and sensitivity analyses to assess robustness [16–18].

Materials and methods

Study design and population

This prospective cohort study utilized data from the English Longitudinal Study of Ageing (ELSA), a nationally representative panel study of community-dwelling adults aged 50 years and older in England [19]. ELSA collects data biennially through face-to-face interviews, self-completion questionnaires, and nurse visits. Biomedical assessments, including fasting blood samples and anthropometric measurements, were obtained during nurse visits conducted at even-numbered waves only (Waves 2, 4, 6, etc.); these data were not collected at odd-numbered waves (Waves 3, 5). The present analysis used data from Wave 2 (2004–2005) as baseline through Wave 6 (2012–2013) for trajectory modeling. Follow-up for incident CVD ascertainment began at Wave 2 and extended through Wave 9 (2018–2019), with events recorded at each subsequent wave. Only exposure data collected

prior to the first reported CVD event were used for trajectory construction; post-event measurements were excluded (see Statistical Analysis for details). From an initial pool of 9432 participants at Wave 2, we excluded individuals aged below 50 years (n = 89), those with missing triglyceride-glucose index (TyG) data (n = 4125), missing body mass index (BMI) data (n = 1058), missing waist-to-height ratio (WHtR) data (n = 868), and those with prevalent CVD at baseline (n = 1484). The final analytic sample comprised 1808 participants free of CVD at baseline. The study population was predominantly White (98.7%), consistent with the demographic composition of ELSA. A comparison of baseline characteristics between included and excluded participants is presented in Supplementary Table 1. This study was conducted in accordance with the principles of the Declaration of Helsinki and followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

Exposure assessment

The TyG index was calculated as ln[fasting triglycerides (mg/dL) × fasting glucose (mg/dL)/2]. BMI was computed as weight in kilograms divided by height in meters squared. WHtR was derived by dividing waist circumference by standing height. All three indicators were measured at Waves 2, 4, and 6, providing three repeated assessments over approximately 8 years.

Group-based multi-trajectory modeling

We employed group-based multi-trajectory modeling (GBMTM) using the flexmix package in R to jointly identify latent trajectory classes over the repeated measurements at Waves 2, 4, and 6. Two parallel modeling approaches were conducted: (1) BMI combined with TyG, and (2) WHtR combined with TyG. For each approach, both indicators were standardized using z-score transformation prior to model fitting. Quadratic polynomial functions of time were specified within each trajectory class, with Gaussian distributions assumed for both outcomes. Models were estimated for K = 2 through K = 6 groups. In addition to quadratic polynomials, linear and cubic time functions were evaluated as a sensitivity analysis. The optimal number of groups was determined by comparing BIC, AIC, entropy, average posterior probability of assignment (AvePP; threshold ≥ 0.80), odds of correct classification (OCC), minimum class proportion, and clinical interpretability. Detailed model fit indices for all candidate solutions are reported in Supplementary Table 4. The four-group solution (K = 4) was selected as optimal for both approaches. Trajectory groups were labeled according to their baseline metabolic-anthropometric profiles in ascending risk order. Individuals were assigned to the trajectory group for which they had the highest posterior probability. Because trajectory group membership is estimated rather than directly observed, posterior probability–weighted Cox models were additionally fitted as a sensitivity analysis to account for classification uncertainty.

Outcome ascertainment

The primary outcome was incident composite CVD, defined as a new diagnosis of heart disease (including heart attack, heart failure, angina, or heart rhythm disorder) or stroke reported during follow-up waves. Secondary outcomes were incident heart disease and incident stroke, analyzed separately. Outcomes were ascertained through self-reported physician diagnoses at each follow-up wave interview. Because exact event dates were not available and events were reported between successive wave interviews, the data are interval-censored in nature. Follow-up time was calculated in months from the Wave 2 interview date to the midpoint between the last event-free wave and the wave at which the event was first reported, or to the end of follow-up (Wave 9), whichever occurred first. The median follow-up time was 11.2 years (interquartile range 8.4–13.6). Participants who died without a CVD event were censored at the time of death.

Covariates

Covariates were selected a priori based on clinical relevance and existing literature [11]. Demographic variables included age (continuous), gender, race/ethnicity (White vs. non-White, based on ELSA harmonized masked race variable), and education level. Lifestyle factors included smoking status and drinking status. Comorbidities included hypertension, diabetes mellitus, hypercholesterolemia, chronic lung disease, arthritis, osteoporosis, and cancer, all ascertained by self-report at baseline. Medication use was recorded from nurse visit medication inventories and included antihypertensive medication, antidiabetic medication (insulin and oral antidiabetic agents), and cholesterol-lowering medication. Cholesterol-lowering medication data were obtained from Wave 3 due to unavailability at Wave 2. Forced vital capacity (FVC), measured in liters during the nurse visit, was included as a proxy for cardiorespiratory fitness. Depressive symptoms were assessed using the eight-item Centre for Epidemiological Studies Depression Scale (CES-D 8), with a score of 3 or above indicating clinically significant depressive symptoms. The proportion of missing data for each covariate is reported in Supplementary Table 2. Missing values for education (8.68%), drinking status (4.81%), forced vital capacity (18.53%), and depressive symptoms (0.17%) were handled using multiple imputation by chained equations (MICE) with 20 imputed datasets, and results were pooled using Rubin's rules. A complete case analysis was conducted as a sensitivity analysis.

Statistical analysis

Baseline characteristics were compared between participants with and without incident CVD, and separately between included and excluded participants, using t-tests or Wilcoxon rank-sum tests for continuous variables and chi-squared or Fisher's exact tests for categorical variables. Standardized mean differences (SMDs) were calculated to quantify between-group imbalances. Baseline characteristics stratified by trajectory group membership are presented in Tables 2 and 3. Cumulative incidence of cardiovascular outcomes across trajectory groups was estimated using the Kaplan–Meier method, with between-group differences assessed by log-rank tests. Cox proportional hazards regression models were used to estimate hazard ratios (HRs) with 95% confidence intervals for the association between trajectory group membership and incident CVD outcomes. Four sequential models were fitted: Model 1 adjusted for age, gender, race/ethnicity, and education level; Model 2 additionally adjusted for smoking status and alcohol consumption; Model 3 further adjusted for hypertension, diabetes mellitus, hypercholesterolemia, chronic lung disease, arthritis, osteoporosis, and depressive symptoms; Model 4 further adjusted for antihypertensive medication, antidiabetic medication, cholesterol-lowering medication, and forced vital capacity. The proportional hazards assumption was assessed for all models using Schoenfeld residual tests, with both individual covariate-level and global tests reported. The lowest-risk trajectory group served as the reference category in all models. Subgroup analyses were performed stratified by age group (< 65 vs. ≥ 65 years), gender, education level, diabetes status, hypertension status, and smoking status, with multiplicative interaction terms tested using likelihood ratio tests. Nine sensitivity analyses were conducted to evaluate the robustness of the primary findings: (1) excluding CVD events occurring within the first 2 years of follow-up to minimize reverse causation; (2) Fine–Gray subdistribution hazard regression treating non-CVD death as a competing event; (3) excluding participants with baseline diabetes or hypertension; (4) complete case analysis without multiple imputation; (5) excluding users of lipid-lowering or glucose-lowering medications to address potential confounding by pharmacological modification of TyG components; (6) posterior probability–weighted Cox models to account for uncertainty in trajectory group assignment; (7) complementary log–log regression models to address interval censoring of event times [20]; (8) re-estimation of trajectory models using alternative time functions (linear and cubic polynomials); and (9) cross-classified analysis based on median-split BMI × TyG and WHtR × TyG categories to assess concordance between trajectory-based and categorical approaches. The Benjamini–Hochberg false discovery rate (FDR) procedure was applied to adjust for multiple comparisons across trajectory groups. Statistical power for each trajectory group comparison was estimated post hoc. Discriminative performance of the BMI + TyG and WHtR + TyG approaches was formally compared using differences in Harrell's C-statistic, the net reclassification improvement (NRI), and the integrated discrimination improvement (IDI). All analyses were performed using R version 4.3.0, and two-sided P values less than 0.05 were considered statistically significant.

Results

Baseline characteristics

A total of 9432 ELSA participants were initially identified at Wave 2. After applying the exclusion criteria, 1808 individuals free of CVD at baseline were included in the final analysis, of whom 263 (14.5%) developed incident CVD during follow-up (Fig. 1). A comparison of baseline characteristics between included (n = 1808) and excluded (n = 7624) participants is presented in Supplementary Table 1. The two groups were broadly similar with respect to age, gender, race/ethnicity, education, smoking status, diabetes, and depression (all SMD < 0.12). Included participants were more likely to report alcohol consumption (94.6% vs. 91.9%, $P < 0.001$) and less likely to have hypertension (29.0% vs. 34.4%, $P < 0.001$) or arthritis (29.8% vs. 33.3%, $P = 0.005$). Importantly, the incidence of CVD was virtually identical between the two groups (14.7% vs. 14.8%, $P = 0.933$, SMD = 0.003), suggesting that missingness was not substantially associated with the primary outcome. The proportion of missing data for each baseline covariate in the analytic sample is reported in Supplementary Table 2; most variables had no missing values, with the exceptions of forced vital capacity (18.53%), education (8.68%), drinking status (4.81%), and depression score (0.17%).

The baseline characteristics of the analytic sample stratified by incident CVD status are presented in Table 1. The study population was predominantly White (98.7%). The median age was comparable between those who did and did not develop CVD (62.00 vs. 60.00 years, $P = 0.401$), and the gender distribution was well balanced (55.5% vs. 57.0% female, $P = 0.712$). Participants who subsequently developed CVD had significantly higher baseline TyG index (median 8.63 vs. 8.55, $P = 0.005$), BMI (median 28.90 vs. 26.89 kg/m^2, $P < 0.001$), and WHtR (median 0.59 vs. 0.56, $P < 0.001$). The prevalence of diabetes mellitus (4.6% vs. 2.1%, $P = 0.034$), hypercholesterolemia (21.7% vs. 15.4%, $P = 0.014$), and the use of oral antidiabetic medication (3.4% vs. 1.3%, $P = 0.023$) and cholesterol-lowering medication (20.5% vs. 15.0%, $P = 0.027$) were higher in the incident CVD group. No significant differences were observed for education, marital status, smoking,

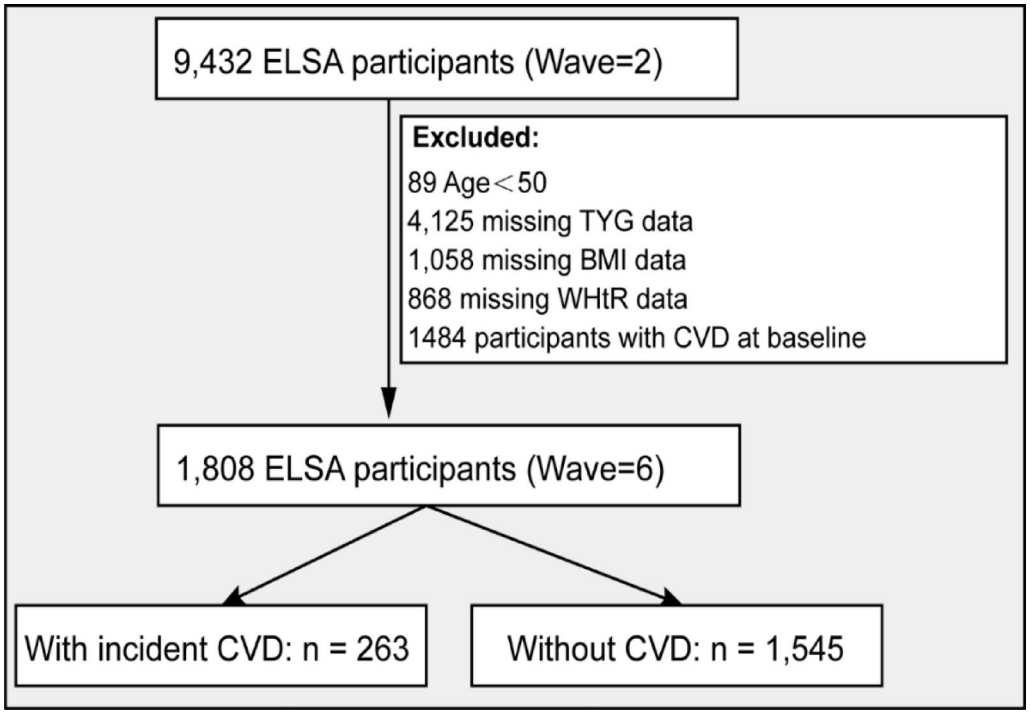


Fig. 1 Flow diagram of participant selection from the English Longitudinal Study of Ageing (ELSA)

drinking, hypertension, forced vital capacity, antihypertensive medication use, or depression.

Temporal distributions of exposure variables

The distributions of TyG index, BMI, and WHtR across the study period are summarized in Supplementary Table 3 and Fig. 2A–C. The mean TyG index remained remarkably stable from Wave 2 (8.61 ± 0.51) to Wave 6 (8.59 ± 0.48), whereas BMI showed a modest upward trend from 27.75 ± 4.73 to 28.14 ± 4.99 kg/m^2. WHtR followed a similar pattern, rising slightly from 0.57 ± 0.07 to 0.58 ± 0.08. These temporal patterns indicated that metabolic status as reflected by TyG remained largely unchanged over the eight-year period, while anthropometric indicators exhibited a gradual increasing tendency.

Identification of joint trajectory groups

GBMTM was applied to identify latent trajectory classes based on the repeated measurements. For the BMI + TyG approach, the four-group solution yielded the lowest AIC (24,235.7) and BIC (24,466.7) among all candidate models (K = 2–6) and was selected as optimal (Fig. 3A). Similarly, for the WHtR + TyG approach, the four-group solution demonstrated the best fit (AIC 25,612.5, BIC 25,843.4; Fig. 3B). Detailed model fit indices for K = 2–6 solutions, including entropy, average posterior probability, odds of correct classification, and minimum class proportion, are provided in Supplementary Table 4. For the selected four-group models, the average posterior probabilities were 0.82 (BMI + TyG) and 0.81 (WHtR + TyG), with entropy values of 0.75 and 0.74, respectively, indicating acceptable classification quality. In the BMI + TyG approach (Fig. 3C), the four groups were labeled as normal weight-low TyG (n = 339, 18.8%), overweight-moderate TyG (n = 625, 34.6%), obese-high TyG (n = 556, 30.7%), and severely obese-high TyG (n = 288, 15.9%). The WHtR + TyG approach (Fig. 3D) yielded non-abdominal obesity-low TyG (n = 211, 11.7%), mild abdominal obesity-moderate TyG (n = 542, 30.0%), moderate abdominal obesity-high TyG (n = 694, 38.4%), and severe abdominal obesity-high TyG (n = 361, 20.0%). All trajectory groups maintained relatively stable levels across waves. The baseline characteristics of participants stratified by trajectory group are presented in Table 2 (BMI + TyG) and Table 3 (WHtR + TyG). In both approaches, higher-risk trajectory groups were characterized by progressively greater prevalence of hypertension, diabetes mellitus, arthritis, and medication use (antihypertensive, antidiabetic, and cholesterol-lowering), as well as higher levels of depression and lower forced vital capacity, whereas education level decreased across ascending groups.

Cumulative incidence and cox regression analyses

Kaplan–Meier analysis revealed significant differences in cumulative incidence across trajectory groups for all outcomes in both approaches (all log-rank $P < 0.001$; Fig. 4A–F). In the BMI + TyG approach, event rates increased

Table 1 Baseline characteristics of the study population stratified by incident cardiovascular disease status

Characteristics	Category	Overall (n = 1808)	Non-CVD (n = 1545)	Incident CVD (n = 263)	*P* value	SMD
Age, years		61.00 [57.00, 67.00]	60.00 [57.00, 67.00]	62.00 [57.00, 67.00]	0.401	0.045
Gender	Female	1026 (56.7)	880 (57.0)	146 (55.5)	0.712	0.029
	Male	782 (43.3)	665 (43.0)	117 (44.5)		
Race/Ethnicity	White	1784 (98.7)	1524 (98.6)	260 (98.9)	1.000	0.020
	Non-white	24 (1.3)	21 (1.4)	3 (1.1)		
Education level	High	327 (18.5)	288 (19.1)	39 (15.1)	0.279	0.111
	Low	492 (27.8)	414 (27.4)	78 (30.2)		
	Medium	948 (53.7)	807 (53.5)	141 (54.7)		
Marital status	Divorced	221 (12.2)	189 (12.2)	32 (12.2)	0.817	0.065
	Married	1349 (74.6)	1154 (74.7)	195 (74.1)		
	Single	75 (4.1)	66 (4.3)	9 (3.4)		
	Widowed	163 (9.0)	136 (8.8)	27 (10.3)		
Smoking status	No	1610 (89.0)	1377 (89.1)	233 (88.6)	0.881	0.017
	Yes	198 (11.0)	168 (10.9)	30 (11.4)		
Drinking status	No	99 (5.6)	78 (5.2)	21 (8.0)	0.085	0.116
	Yes	1672 (94.4)	1432 (94.8)	240 (92.0)		
TyG index		8.57 [8.25, 8.94]	8.55 [8.23, 8.92]	8.63 [8.35, 9.01]	0.005	0.213
Body mass index, kg/m^2		27.03 [24.54, 30.25]	26.89 [24.33, 29.74]	28.90 [25.88, 32.66]	< 0.001	0.452
Waist-to-height ratio		0.56 [0.52, 0.61]	0.56 [0.51, 0.60]	0.59 [0.55, 0.65]	< 0.001	0.506
Forced vital capacity, L		3.36 [2.77, 4.14]	3.36 [2.77, 4.18]	3.28 [2.76, 4.04]	0.389	0.036
Hypertension	No	1263 (69.9)	1091 (70.6)	172 (65.4)	0.103	0.112
	Yes	545 (30.1)	454 (29.4)	91 (34.6)		
Diabetes mellitus	No	1763 (97.5)	1512 (97.9)	251 (95.4)	0.034	0.135
	Yes	45 (2.5)	33 (2.1)	12 (4.6)		
Hypercholesterolemia	No	1513 (83.7)	1307 (84.6)	206 (78.3)	0.014	0.162
	Yes	295 (16.3)	238 (15.4)	57 (21.7)		
Cancer	No	1712 (94.7)	1462 (94.6)	250 (95.1)	0.890	0.019
	Yes	96 (5.3)	83 (5.4)	13 (4.9)		
Chronic lung disease	No	1746 (96.6)	1491 (96.5)	255 (97.0)	0.849	0.025
	Yes	62 (3.4)	54 (3.5)	8 (3.0)		
Arthritis	No	1267 (70.1)	1092 (70.7)	175 (66.5)	0.200	0.089
	Yes	541 (29.9)	453 (29.3)	88 (33.5)		
Osteoporosis	No	1750 (96.8)	1491 (96.5)	259 (98.5)	0.136	0.127
	Yes	58 (3.2)	54 (3.5)	4 (1.5)		
Antihypertensive medication	No	1613 (89.2)	1384 (89.6)	229 (87.1)	0.270	0.078
	Yes	195 (10.8)	161 (10.4)	34 (12.9)		
Antidiabetic medication	No	1778 (98.3)	1524 (98.6)	254 (96.6)	0.031	0.135
	Yes	30 (1.7)	21 (1.4)	9 (3.4)		
Insulin use	No	1805 (99.8)	1542 (99.8)	263 (100.0)	1.000	0.062
	Yes	3 (0.2)	3 (0.2)	0 (0.0)		
Oral antidiabetic medication	No	1779 (98.4)	1525 (98.7)	254 (96.6)	0.023	0.141
	Yes	29 (1.6)	20 (1.3)	9 (3.4)		
Cholesterol-lowering medication	No	1523 (84.2)	1314 (85.0)	209 (79.5)	0.027	0.146
	Yes	285 (15.8)	231 (15.0)	54 (20.5)		
Depression (CES-D 8 ≥ 3)	No	1526 (84.4)	1302 (84.3)	224 (85.2)	0.780	0.025
	Yes	282 (15.6)	243 (15.7)	39 (14.8)		

Values are presented as mean ± SD, median (IQR), or n (%). *P* values were calculated using t-test or Wilcoxon rank-sum test for continuous variables, and chi-squared test for categorical variables. SMD = standardized mean difference. Race was categorized as White vs Non-white based on ELSA harmonized masked race variable. Depression was defined as CES-D 8 score ≥ 3. Cholesterol-lowering medication data were obtained from Wave 3 due to unavailability at Wave 2. CVD composite endpoint includes heart disease, heart attack, heart failure, angina, heart rhythm disorder, and stroke

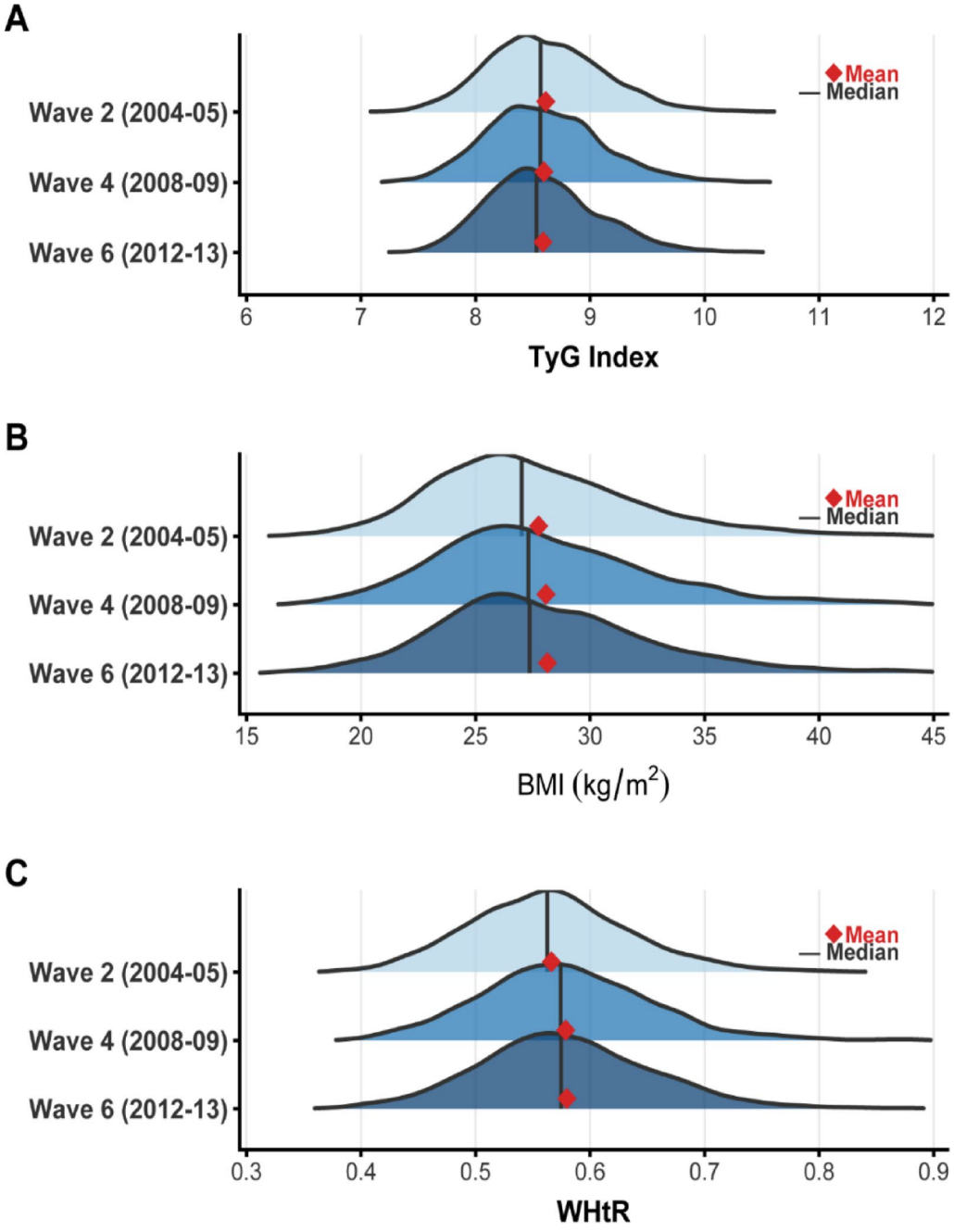


Fig. 2 Ridge density plots showing the distributions of **A** TyG index, **B** BMI, and **C** WHtR across Waves 2, 4, and 6. Red diamonds indicate mean values; vertical lines within distributions represent medians

progressively from the lowest- to the highest-risk group: composite CVD ranged from 5.6% (19/339) in the normal weight–low TyG group to 27.1% (78/288) in the severely obese–high TyG group, heart disease from 5.0 (17/339) to 18.8% (54/288), and stroke from 1.8% (6/339) to 11.8% (34/288) (Table 2). A similar gradient was observed in the WHtR + TyG approach, with composite CVD rates spanning from 5.2 (11/211) to 25.5% (92/361), heart disease from 4.7 (10/211) to 19.7% (71/361), and stroke from 1.9 (4/211) to 9.1% (33/361) (Table 3).

Cox proportional hazards regression analyses confirmed these associations across four progressively adjusted models (Figure 5 and Supplementary Table 5). The global Schoenfeld residual tests indicated no violation of the proportional hazards assumption for any model (all global $P > 0.05$). In the fully adjusted model (Model 4) of the BMI + TyG approach, the risk of

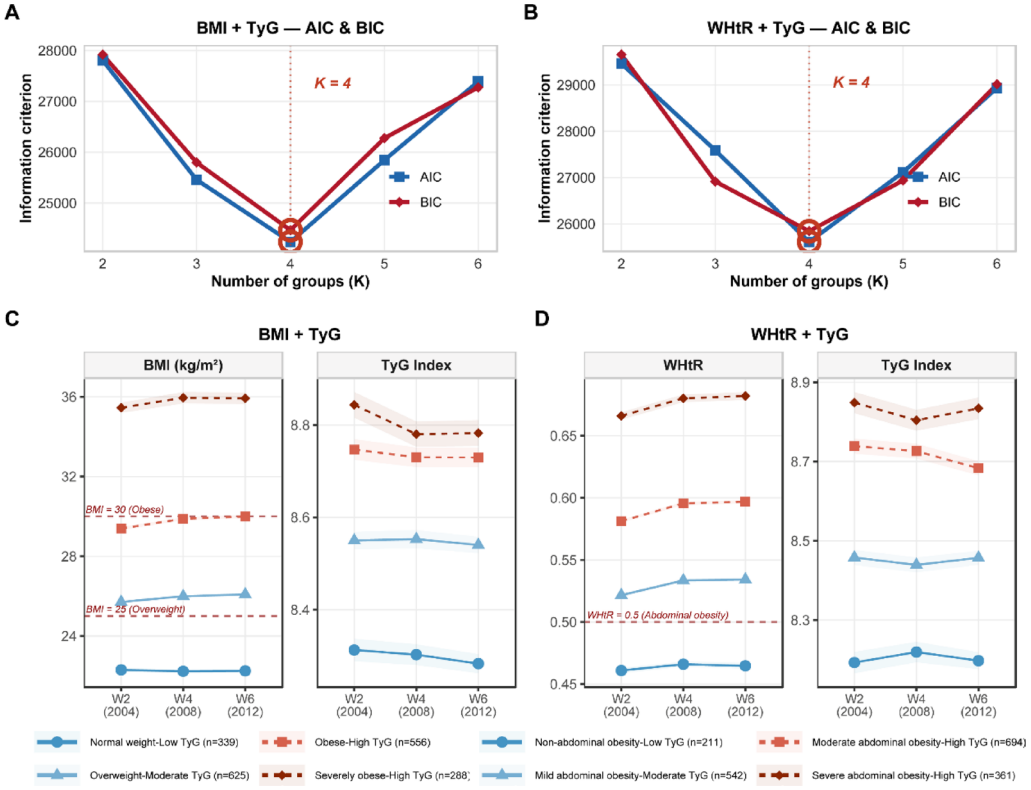


Fig. 3 Model selection and trajectory visualization for group-based multi-trajectory modeling. **A** AIC and BIC values across K = 2–6 groups for BMI + TyG joint trajectories. **B** AIC and BIC values across K = 2–6 groups for WHtR + TyG joint trajectories. **C** Mean trajectories of BMI and TyG index across waves for the four identified trajectory groups (BMI + TyG approach). **D** Mean trajectories of WHtR and TyG index across waves for the four identified trajectory groups (WHtR + TyG approach). Shaded areas represent standard errors. Dashed horizontal lines indicate clinical reference thresholds

composite CVD rose progressively relative to the normal weight–low TyG reference group: overweight–moderate TyG (HR 2.31, 95% CI 1.40–3.83), obese–high TyG (HR 2.72, 95% CI 1.63–4.52), and severely obese–high TyG (HR 5.06, 95% CI 3.01–8.51; all $P \le 0.001$). For heart disease, corresponding HRs were 2.48 (95% CI 1.46–4.22), 2.66 (95% CI 1.55–4.56), and 3.78 (95% CI 2.15–6.65; all $P \le 0.001$). For stroke, only the severely obese–high TyG group reached statistical significance (HR 6.74, 95% CI 2.75–16.54, $P < 0.001$), while the overweight–moderate TyG and obese–high TyG groups showed non-significant elevations (HR 1.81 and 2.06, respectively), consistent with limited statistical power for this endpoint (65.2% and 81.3%, respectively). All associations survived correction for multiple testing using the Benjamini–Hochberg false discovery rate procedure (Fig. 5).

In the WHtR + TyG approach (Model 4), a consistent dose–response pattern emerged for composite CVD, with HRs of 2.22 (95% CI 1.16–4.24), 2.75 (95% CI 1.46–5.18), and 5.12 (95% CI 2.68–9.77) across ascending groups relative to the non-abdominal obesity–low TyG reference. For heart disease, significant associations were observed for mild abdominal obesity–moderate TyG (HR 2.14, 95% CI 1.08–4.22), moderate abdominal obesity–high TyG (HR 2.54, 95% CI 1.30–4.95), and severe abdominal obesity–high TyG (HR 4.10, 95% CI 2.06–8.14). For stroke, only the severe abdominal obesity–high TyG group was statistically significant (HR 5.00, 95% CI 1.71–14.63, $P = 0.003$). These associations remained robust and consistent in direction and magnitude across all four adjustment models (Models 1–4), with minimal attenuation upon sequential addition of covariates including comorbidities and medication use.

Subgroup and sensitivity analyses

Subgroup analyses were conducted across strata of age (< 65 vs. ≥ 65 years), gender, smoking status, education level, hypertension, and diabetes mellitus (Supplementary Table 6). The dose–response gradient between trajectory groups and composite CVD risk was broadly preserved across most strata in both approaches. Interaction tests identified significant effect modification by gender for heart disease in the WHtR + TyG approach (P for interaction = 0.021), and by hypertension for heart disease in the WHtR + TyG approach (P for interaction = 0.020). In the WHtR + TyG approach, a significant interaction by education level was observed for stroke (P for interaction = 0.018). No significant interactions were

Table 2 Baseline characteristics of participants stratified by BMI + TyG joint trajectory groups

Characteristics	Category	Overall (n = 1808)	Normal weight-Low TyG (n = 339)	Overweight-Moderate TyG (n = 625)	Obese-High TyG (n = 556)	Severely obese-High TyG (n = 288)	*P* value	SMD
Age, years		61.00 [57.00, 67.00]	61.00 [56.00, 66.00]	62.00 [57.00, 67.00]	61.00 [57.00, 67.00]	59.00 [55.00, 65.00]	0.001	0.146
Gender	Female	1026 (56.7)	228 (67.3)	318 (50.9)	288 (51.8)	192 (66.7)	< 0.001	0.220
	Male	782 (43.3)	111 (32.7)	307 (49.1)	268 (48.2)	96 (33.3)		
Race/Ethnicity	White	1784 (98.7)	337 (99.4)	615 (98.4)	547 (98.4)	285 (99.0)	0.512	0.058
	Non-white	24 (1.3)	2 (0.6)	10 (1.6)	9 (1.6)	3 (1.0)		
Education level	High	327 (18.5)	87 (26.1)	118 (19.2)	84 (15.5)	38 (13.6)	< 0.001	0.206
	Low	492 (27.8)	71 (21.3)	160 (26.1)	168 (31.0)	93 (33.3)		
	Medium	948 (53.7)	175 (52.6)	335 (54.6)	290 (53.5)	148 (53.0)		
Marital status	Divorced	221 (12.2)	40 (11.8)	78 (12.5)	68 (12.2)	35 (12.2)	0.232	0.142
	Married	1349 (74.6)	245 (72.3)	466 (74.6)	419 (75.4)	219 (76.0)		
	Single	75 (4.1)	22 (6.5)	30 (4.8)	18 (3.2)	5 (1.7)		
	Widowed	163 (9.0)	32 (9.4)	51 (8.2)	51 (9.2)	29 (10.1)		
Smoking status	No	1610 (89.0)	289 (85.3)	562 (89.9)	511 (91.9)	248 (86.1)	0.005	0.125
	Yes	198 (11.0)	50 (14.7)	63 (10.1)	45 (8.1)	40 (13.9)		
Drinking status	No	99 (5.6)	27 (8.0)	23 (3.7)	26 (4.8)	23 (8.2)	0.008	0.117
	Yes	1672 (94.4)	311 (92.0)	591 (96.3)	513 (95.2)	257 (91.8)		
TyG index		8.57 [8.25, 8.94]	8.27 [8.03, 8.59]	8.49 [8.24, 8.87]	8.71 [8.41, 9.05]	8.83 [8.50, 9.11]	< 0.001	0.632
Body mass index, kg/m^2		27.03 [24.54, 30.25]	22.66 [21.23, 23.45]	25.77 [24.81, 26.63]	29.34 [28.26, 30.55]	34.64 [32.80, 37.21]	< 0.001	2.913
Waist-to-height ratio		0.56 [0.52, 0.61]	0.48 [0.46, 0.51]	0.54 [0.51, 0.57]	0.59 [0.57, 0.62]	0.66 [0.63, 0.70]	< 0.001	2.128
Forced vital capacity, L		3.36 [2.77, 4.14]	3.36 [2.82, 4.19]	3.54 [2.85, 4.26]	3.34 [2.80, 4.12]	3.02 [2.52, 3.77]	< 0.001	0.257
Hypertension	No	1263 (69.9)	281 (82.9)	469 (75.0)	350 (62.9)	163 (56.6)	< 0.001	0.340
	Yes	545 (30.1)	58 (17.1)	156 (25.0)	206 (37.1)	125 (43.4)		
Diabetes mellitus	No	1763 (97.5)	338 (99.7)	618 (98.9)	535 (96.2)	272 (94.4)	< 0.001	0.195
	Yes	45 (2.5)	1 (0.3)	7 (1.1)	21 (3.8)	16 (5.6)		
Hypercholesterolemia	No	1513 (83.7)	292 (86.1)	510 (81.6)	467 (84.0)	244 (84.7)	0.290	0.065
	Yes	295 (16.3)	47 (13.9)	115 (18.4)	89 (16.0)	44 (15.3)		
Cancer	No	1712 (94.7)	324 (95.6)	598 (95.7)	515 (92.6)	275 (95.5)	0.078	0.066
	Yes	96 (5.3)	15 (4.4)	27 (4.3)	41 (7.4)	13 (4.5)		
Chronic lung disease	No	1746 (96.6)	324 (95.6)	604 (96.6)	539 (96.9)	279 (96.9)	0.721	0.038
	Yes	62 (3.4)	15 (4.4)	21 (3.4)	17 (3.1)	9 (3.1)		
Arthritis	No	1267 (70.1)	266 (78.5)	446 (71.4)	381 (68.5)	174 (60.4)	< 0.001	0.209
	Yes	541 (29.9)	73 (21.5)	179 (28.6)	175 (31.5)	114 (39.6)		
Osteoporosis	No	1750 (96.8)	320 (94.4)	605 (96.8)	541 (97.3)	284 (98.6)	0.020	0.123
	Yes	58 (3.2)	19 (5.6)	20 (3.2)	15 (2.7)	4 (1.4)		
Antihypertensive medication	No	1613 (89.2)	320 (94.4)	565 (90.4)	487 (87.6)	241 (83.7)	< 0.001	0.190
	Yes	195 (10.8)	19 (5.6)	60 (9.6)	69 (12.4)	47 (16.3)		
Antidiabetic medication	No	1778 (98.3)	339 (100.0)	622 (99.5)	541 (97.3)	276 (95.8)	< 0.001	0.189
	Yes	30 (1.7)	0 (0.0)	3 (0.5)	15 (2.7)	12 (4.2)		
Insulin use	No	1805 (99.8)	339 (100.0)	625 (100.0)	553 (99.5)	288 (100.0)	0.080	0.052
	Yes	3 (0.2)	0 (0.0)	0 (0.0)	3 (0.5)	0 (0.0)		
Oral antidiabetic medication	No	1779 (98.4)	339 (100.0)	622 (99.5)	542 (97.5)	276 (95.8)	< 0.001	0.188
	Yes	29 (1.6)	0 (0.0)	3 (0.5)	14 (2.5)	12 (4.2)		
Cholesterol-lowering medication	No	1523 (84.2)	305 (90.0)	531 (85.0)	460 (82.7)	227 (78.8)	0.001	0.166
	Yes	285 (15.8)	34 (10.0)	94 (15.0)	96 (17.3)	61 (21.2)		
Depression (CES-D 8 ≥ 3)	No	1526 (84.4)	293 (86.4)	536 (85.8)	472 (84.9)	225 (78.1)	0.014	0.114
	Yes	282 (15.6)	46 (13.6)	89 (14.2)	84 (15.1)	63 (21.9)		
Incident CVD	No	1545 (85.5)	320 (94.4)	544 (87.0)	471 (84.7)	210 (72.9)	< 0.001	0.317
	Yes	263 (14.5)	19 (5.6)	81 (13.0)	85 (15.3)	78 (27.1)		
Incident heart disease	No	1585 (87.7)	322 (95.0)	548 (87.7)	481 (86.5)	234 (81.2)	< 0.001	0.225
	Yes	223 (12.3)	17 (5.0)	77 (12.3)	75 (13.5)	54 (18.8)		

Table 2 (continued)

Characteristics	Category	Overall (n = 1808)	Normal weight-Low TyG (n = 339)	Overweight-Moderate TyG (n = 625)	Obese-High TyG (n = 556)	Severely obese-High TyG (n = 288)	*P* value	SMD
Incident stroke	No	1725 (95.4)	333 (98.2)	603 (96.5)	535 (96.2)	254 (88.2)	< 0.001	0.212
	Yes	83 (4.6)	6 (1.8)	22 (3.5)	21 (3.8)	34 (11.8)		

Values are presented as mean ± SD, median (IQR), or n (%). *P* values were calculated using ANOVA or Kruskal–Wallis test for continuous variables, and chi-squared or Fisher's exact test for categorical variables. SMD = standardized mean difference. Race was categorized as White vs Non-white based on ELSA harmonized masked race variable. Depression was defined as CES-D 8 score ≥ 3. Cholesterol-lowering medication data were obtained from Wave 3 due to unavailability at Wave 2. CVD composite endpoint includes heart disease, heart attack, heart failure, angina, heart rhythm disorder, and stroke

detected for age, diabetes, or smoking status in either approach for composite CVD or heart disease. Stroke-specific subgroup analyses were limited by small event numbers in several strata, with many estimates non-estimable, and should be interpreted with caution.

Nine sensitivity analyses consistently corroborated the primary findings (Supplementary Table 7). After excluding early CVD events within the first 2 years, applying Fine–Gray competing risk models, removing participants with baseline diabetes or hypertension, and conducting complete case analysis, the dose–response gradient was preserved, with the highest-risk group retaining HRs ranging from 4.62 to 5.52 for composite CVD across both approaches. Further analyses excluding users of lipid- or glucose-lowering medications yielded similar results (HR 5.38 and 5.48 for the highest-risk groups in the BMI + TyG and WHtR + TyG approaches, respectively). Posterior probability–weighted Cox models, which accounted for uncertainty in trajectory group assignment, produced nearly identical estimates (HR 4.78 and 4.84). Complementary log–log models addressing interval censoring confirmed the robustness of the findings (HR 5.14 and 5.20). Analyses using alternative time functions (linear and cubic polynomials) in the trajectory models yielded comparable results (HR 4.88 and 4.94). Finally, cross-classified analyses based on median-split BMI × TyG and WHtR × TyG categories demonstrated that both high adiposity and high TyG independently contributed to elevated CVD risk, with the combined high–high group showing the strongest associations (HR 3.48 and 3.62 for composite CVD), supporting the concordance between the trajectory-based and categorical approaches. All *P* for trend were < 0.001 across sensitivity analyses.

Formal comparison of discriminative performance between the two trajectory approaches is presented in Supplementary Table 8. The C-statistics for composite CVD were 0.663 (95% CI 0.629–0.697) for the BMI + TyG approach and 0.660 (95% CI 0.626–0.694) for the WHtR + TyG approach, with a non-significant difference of 0.003 ($P = 0.723$). Similar patterns were observed for heart disease (0.652 vs. 0.650, $P = 0.811$) and stroke (0.704 vs. 0.699, $P = 0.685$). The NRI and IDI comparisons also yielded no statistically significant differences between the two approaches for any endpoint (all $P > 0.8$).

Discussion

Summary of principal findings

In this prospective cohort study of 1808 middle-aged and older adults from the English Longitudinal Study of Ageing, we employed group-based multi-trajectory modeling to jointly identify four distinct longitudinal trajectory groups of TyG combined with BMI and TyG combined with WHtR over approximately 8 years. Our primary finding was that membership in higher-risk trajectory groups was associated with progressively elevated risks of incident composite CVD, heart disease, and stroke in a clear dose–response manner. In the fully adjusted BMI + TyG models (Model 4), the severely obese–high TyG group exhibited a 5.06-fold increased risk of composite CVD compared with the normal weight–low TyG reference group, while the corresponding highest-risk group in the WHtR + TyG approach showed a 5.12-fold elevation. For stroke specifically, only the highest-risk trajectory groups reached statistical significance in both approaches, with hazard ratios of 6.74 and 5.00, respectively, consistent with limited statistical power for this endpoint. Subgroup analyses revealed significant effect modification by gender and hypertension status for heart disease in the WHtR + TyG approach, and by education level for stroke. Formal comparison of discriminative performance showed no significant difference between the BMI + TyG and WHtR + TyG approaches (C-statistic difference 0.003, $P = 0.723$ for composite CVD). All nine sensitivity analyses consistently corroborated the primary findings, with hazard ratios for the highest-risk groups ranging from 4.62 to 5.62 for composite CVD across both approaches.

Comparison with cross-sectional and single-timepoint evidence

Our findings are broadly consistent with the growing body of evidence linking TyG-related composite indices to cardiovascular outcomes. In a cross-sectional analysis of 11,937 NHANES participants, Dang et al. reported that TyG-WC and TyG-WHtR demonstrated superior diagnostic efficacy for CVD compared with TyG alone, with TyG-WHtR being the strongest predictor of CVD mortality (HR 1.66, 95% CI 1.21–2.29) [5]. Similarly, a UK Biobank study of 97,331 MASLD patients found that

Table 3 Baseline characteristics of participants stratified by WHtR + TyG joint trajectory groups

Characteristics	Category	Overall (n = 1808)	Non-abdominal obesity-Low TyG (n = 211)	Mild abdominal obesity-Moderate TyG (n = 542)	Moderate abdominal obesity-High TyG (n = 694)	Severe abdominal obesity-High TyG (n = 361)	*P* value	SMD
Age, years		61.00 [57.00, 67.00]	60.00 [56.00, 65.00]	61.00 [57.00, 66.00]	61.00 [57.00, 67.00]	60.00 [56.00, 67.00]	0.022	0.133
Gender	Female	1026 (56.7)	164 (77.7)	309 (57.0)	341 (49.1)	212 (58.7)	< 0.001	0.313
	Male	782 (43.3)	47 (22.3)	233 (43.0)	353 (50.9)	149 (41.3)		
Race/Ethnicity	White	1784 (98.7)	210 (99.5)	533 (98.3)	684 (98.6)	357 (98.9)	0.608	0.064
	Non-white	24 (1.3)	1 (0.5)	9 (1.7)	10 (1.4)	4 (1.1)		
Education level	High	327 (18.5)	62 (30.4)	110 (20.6)	113 (16.7)	42 (12.0)	< 0.001	0.291
	Low	492 (27.8)	39 (19.1)	128 (23.9)	199 (29.4)	126 (35.9)		
	Medium	948 (53.7)	103 (50.5)	297 (55.5)	365 (53.9)	183 (52.1)		
Marital status	Divorced	221 (12.2)	27 (12.8)	69 (12.7)	76 (11.0)	49 (13.6)	0.295	0.133
	Married	1349 (74.6)	150 (71.1)	398 (73.4)	531 (76.5)	270 (74.8)		
	Single	75 (4.1)	12 (5.7)	30 (5.5)	25 (3.6)	8 (2.2)		
	Widowed	163 (9.0)	22 (10.4)	45 (8.3)	62 (8.9)	34 (9.4)		
Smoking status	No	1610 (89.0)	187 (88.6)	484 (89.3)	630 (90.8)	309 (85.6)	0.085	0.084
	Yes	198 (11.0)	24 (11.4)	58 (10.7)	64 (9.2)	52 (14.4)		
Drinking status	No	99 (5.6)	19 (9.1)	21 (3.9)	33 (4.8)	26 (7.5)	0.014	0.124
	Yes	1672 (94.4)	190 (90.9)	512 (96.1)	648 (95.2)	322 (92.5)		
TyG index		8.57 [8.25, 8.94]	8.16 [7.95, 8.39]	8.43 [8.16, 8.75]	8.72 [8.41, 9.02]	8.83 [8.49, 9.14]	< 0.001	0.831
Body mass index, kg/m^2		27.03 [24.54, 30.25]	22.39 [20.82, 23.45]	24.99 [23.55, 26.25]	28.19 [26.55, 30.00]	33.26 [30.82, 36.31]	< 0.001	2.178
Waist-to-height ratio		0.56 [0.52, 0.61]	0.46 [0.44, 0.48]	0.52 [0.50, 0.54]	0.58 [0.56, 0.60]	0.66 [0.63, 0.69]	< 0.001	3.016
Forced vital capacity, L		3.36 [2.77, 4.14]	3.31 [2.84, 4.04]	3.47 [2.88, 4.36]	3.44 [2.82, 4.15]	3.00 [2.50, 3.80]	< 0.001	0.240
Hypertension	No	1263 (69.9)	179 (84.8)	421 (77.7)	468 (67.4)	195 (54.0)	< 0.001	0.389
	Yes	545 (30.1)	32 (15.2)	121 (22.3)	226 (32.6)	166 (46.0)		
Diabetes mellitus	No	1763 (97.5)	211 (100.0)	534 (98.5)	679 (97.8)	339 (93.9)	< 0.001	0.206
	Yes	45 (2.5)	0 (0.0)	8 (1.5)	15 (2.2)	22 (6.1)		
Hypercholesterolemia	No	1513 (83.7)	184 (87.2)	452 (83.4)	574 (82.7)	303 (83.9)	0.485	0.065
	Yes	295 (16.3)	27 (12.8)	90 (16.6)	120 (17.3)	58 (16.1)		
Cancer	No	1712 (94.7)	198 (93.8)	520 (95.9)	653 (94.1)	341 (94.5)	0.471	0.050
	Yes	96 (5.3)	13 (6.2)	22 (4.1)	41 (5.9)	20 (5.5)		
Chronic lung disease	No	1746 (96.6)	203 (96.2)	525 (96.9)	674 (97.1)	344 (95.3)	0.451	0.054
	Yes	62 (3.4)	8 (3.8)	17 (3.1)	20 (2.9)	17 (4.7)		
Arthritis	No	1267 (70.1)	168 (79.6)	396 (73.1)	487 (70.2)	216 (59.8)	< 0.001	0.230
	Yes	541 (29.9)	43 (20.4)	146 (26.9)	207 (29.8)	145 (40.2)		
Osteoporosis	No	1750 (96.8)	199 (94.3)	516 (95.2)	680 (98.0)	355 (98.3)	0.002	0.134
	Yes	58 (3.2)	12 (5.7)	26 (4.8)	14 (2.0)	6 (1.7)		
Antihypertensive medication	No	1613 (89.2)	196 (92.9)	506 (93.4)	608 (87.6)	303 (83.9)	< 0.001	0.180
	Yes	195 (10.8)	15 (7.1)	36 (6.6)	86 (12.4)	58 (16.1)		
Antidiabetic medication	No	1778 (98.3)	211 (100.0)	539 (99.4)	683 (98.4)	345 (95.6)	< 0.001	0.185
	Yes	30 (1.7)	0 (0.0)	3 (0.6)	11 (1.6)	16 (4.4)		
Insulin use	No	1805 (99.8)	211 (100.0)	542 (100.0)	693 (99.9)	359 (99.4)	0.208	0.065
	Yes	3 (0.2)	0 (0.0)	0 (0.0)	1 (0.1)	2 (0.6)		
Oral antidiabetic medication	No	1779 (98.4)	211 (100.0)	539 (99.4)	684 (98.6)	345 (95.6)	< 0.001	0.183
	Yes	29 (1.6)	0 (0.0)	3 (0.6)	10 (1.4)	16 (4.4)		
Cholesterol-lowering medication	No	1523 (84.2)	194 (91.9)	469 (86.5)	581 (83.7)	279 (77.3)	< 0.001	0.221
	Yes	285 (15.8)	17 (8.1)	73 (13.5)	113 (16.3)	82 (22.7)		
Depression (CES-D 8 ≥ 3)	No	1526 (84.4)	182 (86.3)	466 (86.0)	591 (85.2)	287 (79.5)	0.037	0.094
	Yes	282 (15.6)	29 (13.7)	76 (14.0)	103 (14.8)	74 (20.5)		
Incident CVD	No	1545 (85.5)	200 (94.8)	481 (88.7)	595 (85.7)	269 (74.5)	< 0.001	0.311
	Yes	263 (14.5)	11 (5.2)	61 (11.3)	99 (14.3)	92 (25.5)		

Table 3 (continued)

Characteristics	Category	Overall (n = 1808)	Non-abdominal obesity-Low TyG (n = 211)	Mild abdominal obesity-Moderate TyG (n = 542)	Moderate abdominal obesity-High TyG (n = 694)	Severe abdominal obesity-High TyG (n = 361)	*P* value	SMD
Incident heart disease	No	1585 (87.7)	201 (95.3)	487 (89.9)	607 (87.5)	290 (80.3)	< 0.001	0.249
	Yes	223 (12.3)	10 (4.7)	55 (10.1)	87 (12.5)	71 (19.7)		
Incident stroke	No	1725 (95.4)	207 (98.1)	525 (96.9)	665 (95.8)	328 (90.9)	< 0.001	0.174
	Yes	83 (4.6)	4 (1.9)	17 (3.1)	29 (4.2)	33 (9.1)		

Values are presented as mean ± SD, median (IQR), or n (%). *P* values were calculated using ANOVA or Kruskal–Wallis test for continuous variables, and chi-squared or Fisher's exact test for categorical variables. SMD = standardized mean difference. Race was categorized as White vs Non-white based on ELSA harmonized masked race variable. Depression was defined as CES-D 8 score ≥ 3. Cholesterol-lowering medication data were obtained from Wave 3 due to unavailability at Wave 2. CVD composite endpoint includes heart disease, heart attack, heart failure, angina, heart rhythm disorder, and stroke

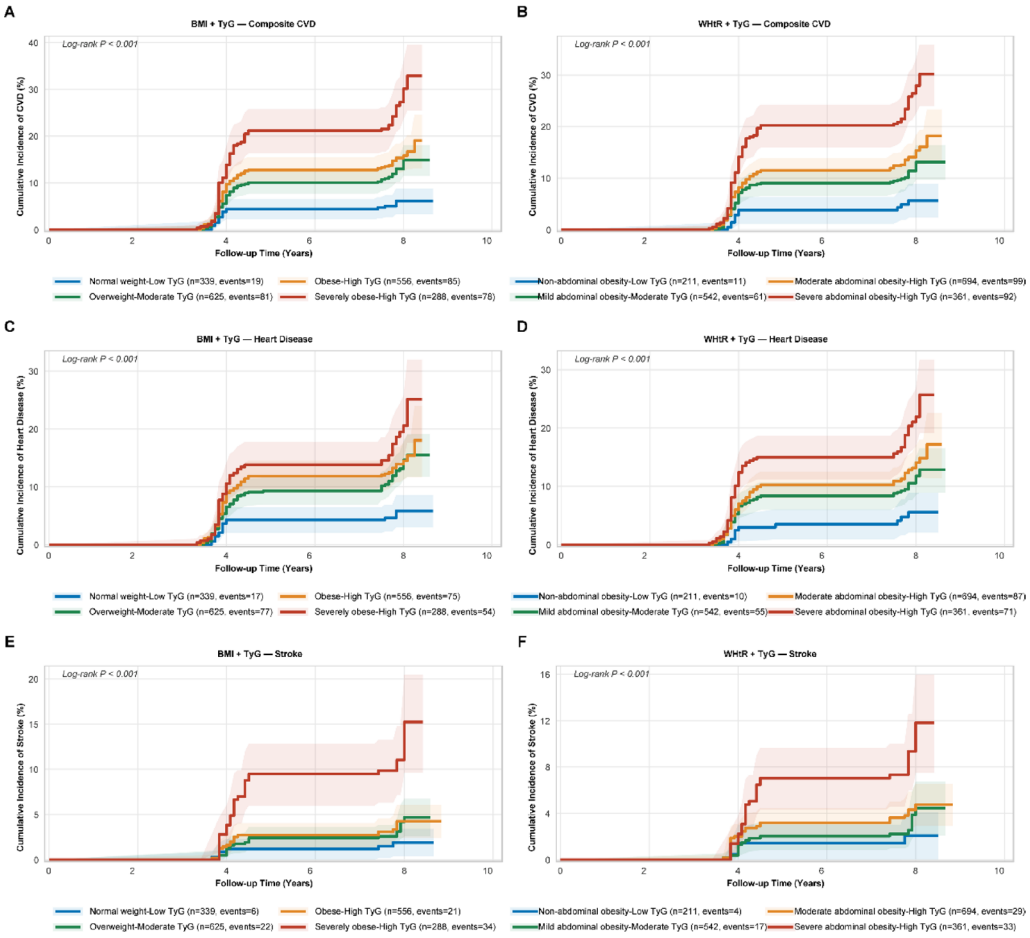


Fig. 4 Cumulative incidence curves of cardiovascular outcomes by trajectory group. Left column: BMI + TyG trajectory groups; right column: WHtR + TyG trajectory groups. Rows from top to bottom: **A**, **B** composite CVD, **C**, **D** heart disease, **E**, **F** stroke. Shaded areas represent 95% confidence intervals. P values were derived from log-rank tests

TyG-WHtR yielded the highest hazard ratio for overall CVD incidence at 1.39 when comparing the highest to lowest quartiles, outperforming TyG-BMI and TyG-WC [12]. A meta-analysis by Rao et al. involving 871,728 subjects confirmed that the highest TyG-BMI category was associated with a 62% increased CVD risk compared to the lowest category [18]. More recently, the most comprehensive meta-analysis to date, encompassing 85 studies, demonstrated that per 1-SD increase, TyG-WHtR and TyG-WC generally conferred greater risk increments than TyG-BMI for CVD incidence and cardiovascular mortality [7]. Our study extends these findings by demonstrating that the magnitude of association is substantially amplified when longitudinal trajectory patterns rather than single baseline measurements are considered, with hazard ratios in the highest-risk

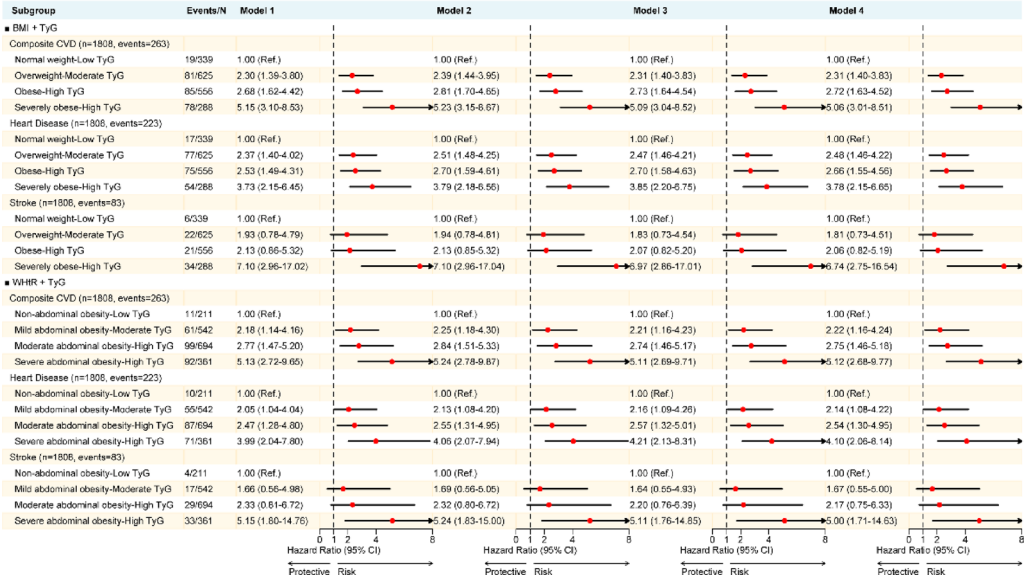


Fig. 5 Forest plots of hazard ratios (95% CI) for the association between joint trajectory groups and incident cardiovascular outcomes. Results are presented for both BMI + TyG and WHtR + TyG approaches across three adjustment levels: Model 1 (age, gender, race/ethnicity, education), Model 2 (Model 1 + smoking, drinking), Model 3 (Model 2 + hypertension, diabetes mellitus, hypercholesterolemia, chronic lung disease, arthritis, osteoporosis, depressive symptoms), and Model 4 (Model 3 + antihypertensive medication, antidiabetic medication, cholesterol-lowering medication, forced vital capacity). The lowest-risk trajectory group served as the reference category

groups reaching 5.06–5.12, considerably exceeding those reported in cross-sectional or single-timepoint analyses.

Longitudinal trajectory approaches and added predictive value

A key contribution of our study lies in the application of joint multi-trajectory modeling, which captures the heterogeneity in concurrent metabolic and anthropometric changes over time. This approach aligns with emerging evidence that longitudinal assessment of TyG-related indices enhances CVD prediction beyond static baseline values. Wang et al., using CHARLS data from 3505 participants, demonstrated that longitudinal trajectories of TyG combined with obesity indices, particularly TyG-WC and TyG-CVAI, exhibited superior CVD risk assessment performance compared with baseline measurements alone [9]. Feng et al. identified four distinct TyG-BMI trajectories and found that individuals in the high-stable group had 1.694-fold higher CVD odds than those in the low-stable group using longitudinal targeted maximum likelihood estimation [10]. Zhu et al. similarly reported that changes in TyG-WC showed the strongest association with incident CVD, with progressive risk elevation across K-means-derived trajectory classes [8]. Furthermore, Ren et al. found that participants with high and slowly increasing TyG-WHtR trajectories had significantly elevated CVD risk (OR 1.58) compared to those with consistently low levels [15]. Zeng et al. confirmed that cumulative average TyG-WHtR independently predicted incident CVD with an 18.3% higher risk per 1-SD increase [21]. Our study advances this literature by simultaneously comparing BMI-based and WHtR-based trajectory approaches within the same cohort. Formal discriminative comparison using C-statistics, NRI, and IDI revealed no statistically significant differences between the two approaches for any cardiovascular endpoint (all $P > 0.68$), indicating that the two approaches provide comparable predictive performance in this population of older adults.

Differential performance of BMI-based versus WHtR-based composite indices

The parallel comparison of BMI- and WHtR-based approaches represents a notable aspect of our analysis. Both approaches identified robust dose–response associations of comparable magnitude (Model 4 HRs for composite CVD: 5.06 vs. 5.12 for the highest-risk groups), and formal discriminative comparisons confirmed no statistically significant differences in predictive performance. This equivalence may reflect the distinct but complementary pathophysiological dimensions captured by these anthropometric measures. WHtR is recognized as a superior indicator of central adiposity compared with BMI, which cannot differentiate between lean and fat mass [22]. Park et al. reported in the Ansung-Ansan cohort that TyG-WC and TyG-WHtR showed significantly higher C-indices than TyG-BMI for predicting new-onset CVD in non-diabetic participants, with TyG-WC achieving the highest hazard ratio of 2.37 in the highest tertile [23]. In hypertensive populations, Li et al.

found TyG-WHtR exhibited the strongest mortality associations, with each 1-unit increase linked to 41.7% and 48.1% higher risks of all-cause and cardiovascular mortality, respectively [6]. He et al. confirmed that among MetS patients, TyG-WC demonstrated the strongest associations with total CVD incidence (HR 1.42 for highest vs. lowest quartile) and provided the greatest incremental predictive value [17]. However, it is noteworthy that some studies have reported context-dependent variations. For instance, in COPD patients, TyG-BMI showed no significant association with all-cause mortality whereas TyG-WHtR demonstrated the strongest and most consistent predictive value [24]. Additionally, among diabetic patients, Xiao et al. observed a U-shaped relationship between TyG-BMI and mortality, suggesting that the prognostic utility of BMI-based indices may vary across metabolic subpopulations [25]. The recently proposed TyG-WWI has also shown promising predictive capacity for cardiovascular mortality, achieving the highest AUC among TyG-related indices in NHANES data [14, 26]. These observations collectively support the notion that central obesity indicators integrated with metabolic markers may provide more nuanced cardiovascular risk stratification than BMI-based composites alone.

Nevertheless, it is important to note that in older adults, the interpretation of BMI is complicated by age-related sarcopenia. BMI tends to decline after approximately age 65 due to loss of lean muscle mass, even as central and visceral adiposity increases. In our cohort, with a baseline age of ≥ 50 years and 8 years of follow-up, a substantial proportion of participants crossed the age-65 threshold during the trajectory observation period. Age-driven BMI decline could therefore be misinterpreted by the trajectory model as a favorable metabolic trajectory, and the normal weight–low TyG reference group may disproportionately include older individuals whose BMI fell due to sarcopenia rather than genuine metabolic improvement. WHtR is less susceptible to this age-related artifact because waist circumference tends to increase with age even as BMI falls, reflecting central fat redistribution. This differential susceptibility to sarcopenic misclassification may partly explain why the two approaches yielded similar rather than divergent discriminative performance in our older cohort, and should be considered when interpreting BMI-based trajectory findings in aging populations. The age-stratified subgroup analyses showed no statistically significant interaction by age group in either approach (all P for interaction > 0.2), though the point estimates were somewhat attenuated in participants aged ≥ 65 years, which is consistent with this hypothesis.

Potential mechanisms and population-specific considerations

The biological plausibility of our findings is supported by converging mechanistic evidence. Insulin resistance, as reflected by the TyG index, promotes endothelial dysfunction, chronic inflammation, and atherogenic dyslipidemia, while central obesity further amplifies these pathways through visceral adipose tissue-mediated secretion of pro-inflammatory cytokines and adipokines [27]. Hu et al. provided novel mechanistic insights by demonstrating that biological age acceleration partially mediates the associations between TyG-related obesity indices and incident CVD, with mediation proportions ranging from 10.03 to 25.46% [22]. The significant effect modifications observed in our study, including by gender and hypertension status for heart disease in the WHtR + TyG approach, are biologically plausible. Sex differences in fat distribution and hormonal regulation of insulin sensitivity may contribute to differential associations, while the interaction with hypertension status may reflect the heterogeneous metabolic effects of different antihypertensive drug classes on TyG components, as both thiazide diuretics and beta-blockers can raise triglycerides and glucose levels whereas ACE inhibitors and ARBs have more neutral metabolic profiles. Notably, our findings in a European cohort complement the predominantly Asian evidence base. Most longitudinal TyG trajectory studies have utilized Chinese cohorts such as CHARLS [13, 28, 29], and cross-sectional evidence has largely derived from NHANES [30–33]. Evidence from UK populations has primarily focused on baseline TyG measurements in the UK Biobank [34, 35]. Our use of ELSA data provides important validation that longitudinal TyG-obesity trajectory associations with CVD are generalizable to older European populations.

Strengths and limitations

This study has several strengths. First, we employed group-based multi-trajectory modeling with three repeated measurements over 8 years, enabling the capture of joint longitudinal dynamics of metabolic and anthropometric indices that single-timepoint analyses cannot achieve. The classification quality of the selected four-group models was supported by average posterior probabilities of 0.82 and 0.81, with entropy values of 0.75 and 0.74, and posterior probability–weighted Cox models yielded nearly identical results, confirming that uncertainty in group assignment did not materially affect the findings. Second, the parallel comparison of BMI-based and WHtR-based approaches within the same cohort, supported by formal discriminative metrics (C-statistics, NRI, IDI), provides direct quantitative evidence on the comparative utility of these composites. Third, comprehensive adjustment across four sequential models

incorporating demographics, lifestyle, comorbidities, and medication use, coupled with nine robust sensitivity analyses including competing risk models, interval-censored models, exclusion of early events, exclusion of medication users, posterior probability–weighted models, alternative time functions, and cross-classified analyses, strengthens confidence in the observed associations. Fourth, the ELSA cohort provides a nationally representative sample of older English adults, enhancing generalizability to European populations. However, several limitations merit consideration. The final analytic sample of 1808 participants represents approximately 19% of the initial Wave 2 cohort, with the largest exclusion due to missing fasting blood data required for TyG calculation (n = 4125), which was only collected during nurse visits and is unlikely to be missing completely at random. However, a formal comparison of baseline characteristics between included and excluded participants showed no significant differences in age, gender, race/ethnicity, education, diabetes, or depression, and importantly, the CVD incidence was virtually identical between groups (14.7% vs. 14.8%, $P = 0.933$), suggesting that selection bias did not substantially affect the primary outcome. Statistical power was limited for stroke analyses, where only the highest-risk groups reached significance (post hoc power estimates of 65–85% for intermediate groups). CVD outcomes were ascertained through self-reported physician diagnoses rather than adjudicated medical records or registry linkage, which may lead to outcome misclassification. Additionally, the absence of exact event dates means that the data are interval-censored between successive wave interviews; however, complementary log–log models explicitly accounting for this interval censoring produced results consistent with the standard Cox analyses. The trajectory modeling utilized only three timepoints at even-numbered waves, which constrains the ability to detect complex non-linear temporal patterns. Notably, the identified trajectories remained relatively stable across waves, suggesting that the groups may primarily reflect different baseline risk strata rather than meaningful longitudinal evolution. However, this stability is consistent with the modest changes in BMI, WHtR, and TyG expected in this age group over an eight-year period, and cross-classified analyses based on baseline median-split categories yielded concordant results, suggesting that the trajectory approach captures the same underlying risk stratification through a more robust longitudinal framework. Additionally, although Model 4 adjusted for antihypertensive, antidiabetic, and cholesterol-lowering medication use, and a sensitivity analysis excluding lipid- and glucose-lowering medication users confirmed the robustness of the findings, residual confounding from unmeasured factors such as habitual physical activity, cardiorespiratory fitness beyond forced vital capacity, dietary patterns, and heterogeneity in antihypertensive drug classes cannot be excluded. Furthermore, because this is an observational study employing latent-class methodology, the associations reported should be interpreted as predictive rather than causal.

Clinical implications and future directions

Our findings carry potential implications for cardiovascular risk stratification in aging populations, though several caveats apply. The demonstration that joint metabolic-anthropometric trajectories substantially outperform single-timepoint assessments in predicting CVD events supports the integration of serial monitoring of TyG-related composite indices into primary prevention strategies [1–4]. However, it should be acknowledged that trajectory membership can only be determined retrospectively after repeated measurements over several years, and the clinical applicability of this approach would require the development and external validation of prospective prediction algorithms that can estimate trajectory membership from early measurements. The identification of distinct trajectory groups with markedly different CVD risks provides a framework for targeted interventions, particularly for individuals exhibiting persistently elevated or increasing TyG-obesity trajectories. Future research should validate these trajectory-based risk models in larger, more diverse prospective cohorts with longer follow-up, incorporate objective CVD ascertainment through medical records or registry linkage, and explore whether trajectory-guided interventions can modify cardiovascular risk. Additionally, emerging composite indices such as TyG-WWI[14]and TyG combined with novel obesity markers like CVAI and BRI [9, 36, 37] warrant investigation within the multi-trajectory framework to determine optimal composite indices for clinical risk prediction. External validation in larger, ethnically diverse cohorts with objective CVD ascertainment through medical records or registry linkage, along with formal comparison against established CVD risk prediction tools, is needed before clinical implementation can be recommended.

Conclusions

In this prospective cohort study of middle-aged and older English adults, group-based multi-trajectory modeling identified four distinct joint trajectory groups of TyG combined with BMI and TyG combined with WHtR, each demonstrating progressively elevated risks of incident composite CVD, heart disease, and stroke in a dose–response fashion. The highest-risk trajectory groups exhibited approximately five-fold to seven-fold increased hazard ratios for cardiovascular events compared with the lowest-risk groups. Formal discriminative comparison showed no significant difference in

predictive performance between the two approaches. Both BMI-based and WHtR-based approaches yielded comparable predictive associations, and these findings remained robust across nine sensitivity analyses and multiple subgroup analyses. These results suggest that longitudinal joint trajectory assessment of metabolic and anthropometric indices may serve as a valuable tool for identifying individuals at elevated cardiovascular risk. External validation and development of prospective prediction algorithms are warranted before incorporation into clinical cardiovascular risk stratification strategies for older adults.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2: Study timeline illustrating the temporal relationship between exposure assessment, trajectory construction, and outcome follow-up. Biomedical assessments including fasting blood samples and anthropometric measurements (TyG index, BMI, and WHtR) were collected during nurse visits at even-numbered waves only (Waves 2, 4, and 6); no biomedical data were collected at odd-numbered waves (Waves 3 and 5). Group-based multi-trajectory modeling (GBMTM) was applied using exposure data from Waves 2, 4, and 6 (approximately 2004–2013) to construct joint longitudinal trajectories, with only pre-event exposure data used for trajectory classification. Follow-up for incident cardiovascular disease (CVD) ascertainment began at the Wave 2 interview date (time zero) and extended through Wave 9 (2018–2019), with events recorded at each subsequent wave. The median follow-up time was 11.2 years (interquartile range 8.4–13.6). TyG, triglyceride-glucose; BMI, body mass index; WHtR, waist-to-height ratio; GBMTM, group-based multi-trajectory modeling; CVD, cardiovascular disease; IQR, interquartile range.

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Author contributions
ZL, QL, BW, YZ, and YD contributed equally to this work. ZL and QL conceived and designed the study. BW and YZ performed the data analysis and statistical modeling. YD contributed to data interpretation. BT assisted with methodology and critical revision. HY and SY supervised the study, provided critical intellectual input, and served as corresponding authors. ZL drafted the original manuscript. All authors reviewed, revised, and approved the final manuscript.

Data availability
The datasets used and analysed during the current study are publicly available from the UK Data Service (https://ukdataservice.ac.uk/). Access to ELSA data requires registration and approval through the UK Data Service.

Declarations

Ethics approval and consent to participate
The English Longitudinal Study of Ageing (ELSA) was approved by the London Multicentre Research Ethics Committee (MREC/01/2/91) and informed consent was obtained from all participants. This study was conducted in accordance with the principles of the Declaration of Helsinki.

Competing interests
The authors declare that they have no competing interests.

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